

Optimal (Tangency) Portfolio

Folder: Optimal_portfolio | Mean-variance frontier, Capital Market Line and rolling-window animation

1. Objective

Visualise how the mean-variance efficient frontier and tangency portfolio evolve through time as estimates of expected returns and covariances are updated on a rolling window. The folder holds `optimal_portfolio_claude.ipynb` and the resulting animation `portfolio.gif`.

2. Data

- $N = 20$ synthetic assets over $T = 2000$ days; daily expected returns drawn uniformly from 0.03-0.08% and volatilities from 1.2-2.0%.
- Log returns are drawn as independent normals (GBM assumption), so the true asset correlations are zero and variation in the estimated moments comes from sampling noise.
- Risk-free rate 2% per year, converted to a daily rate (divided by 252).

3. Method

- **Rolling estimates:** a 1000-day window slides forward one day at a time (1001 windows) and the sample mean vector and covariance matrix are recomputed in each.
- **Random long-only portfolios:** 5000 weight vectors are sampled from a flat Dirichlet distribution (non-negative, summing to one) and plotted in risk-return space.
- **Analytic efficient frontier:** the unconstrained minimum-variance frontier is computed in closed form with the standard A, B, C, D scalars built from the inverse covariance matrix, $\sigma(m) = \sqrt{(A m^2 - 2 B m + C)/D}$, and the extreme portfolio weights are annotated.
- **Tangency portfolio:** the random portfolio with the highest Sharpe ratio $(m - r_f)/\sigma$ is selected and the Capital Market Line drawn through it.
- **Security Market Line:** betas of each asset to the tangency portfolio, $\text{cov}(R_i, R_T)/\sigma_T^2$, give the right-hand panel.

```
CML: R_p = R_0 + (R_T - R_0)/sigma_T * sigma_p SML: R_i = R_0 + (R_T - R_0) *
cov(R_i, R_T)/sigma_T^2
```

4. Output

A two-panel frame (efficient frontier with CML on the left, SML on the right) is rendered every fourth window and written as a GIF with Pillow, showing the frontier and tangency point wobbling as the window moves.

5. Files

File	Description
optimal_portfolio_claude.ipynb	Data simulation, rolling statistics, frontier maths, animation code
portfolio.gif	Rolling-window animation of frontier, CML and SML